

Shower Device

Happy Campers

Element K: Reflection on the Design

Dormitory residents and dormitory guests of our boarding school need a device to place their shower products. This is because they would want their shower products to be easily accessible in the communal shower. They would also like to do this without having a personal shower caddy and without setting it on the floor because it is unsanitary. The main stakeholders are the dormitory residents. They need a solution because the showers in the dormitory were not built with a solution to this problem and dormitory residents are not permitted to leave their shower products in the shower area. The only place to put a device in the dormitory showers are on a relatively narrow divider wall of less than an inch wide.

Our project involved multiple steps. We identified our problem and design requirements, brainstormed, constructed a prototype, tested our prototype, redesigned a new prototype. Our problem and design requirements were identified by a stakeholder that asked us to create a solution. We also acquired requirements from other involved stakeholders to base a solution design on. Next, we brainstormed multiple ideas based on the requirements given to us. We selected one solution that fit the requirements the best and also was a better solution than any current solution that was in practice. After we selected a solution, we constructed a prototype using a 3D printer. During our testing phase where we saw if the prototype performed at the expected level of the requirements, we identified aspects that could be improved or altered. These parts of the first prototype were used to design and construct a second prototype that aligns with the design requirements more closely.

This justification of the problem helped set up for making a solution a reality. Also, this narrowed down thoughts of solutions to a few ideas based on the requirements listed. We selected our idea not only based on our physical design requirements but also its cost in the form of monetary expenditure, labor, and time. The testing process was very straightforward and gave us the information needed to be confident in our prototype.

During the brainstorming process, it is very easy to try to judge your ideas when they are thought of and then dismissed before they are illustrated or described on a page. Instead of dismissing ideas, write all of them down not only because they could be thought of differently from person to person, but also ideas can be changed or combined in some form before they are materialized.